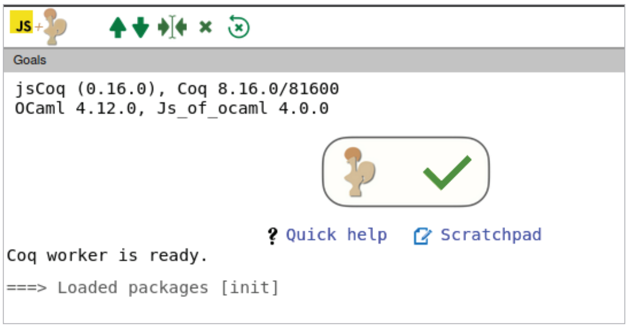

Figure 3: The terminal of the Coq online environment

otherwise, you need to install Emacs on your system. Emacs, Proof General, and Coq can be installed on your system by executing the following three commands in the terminal:

```
sudo apt install emacs
sudo apt install proofgeneral
sudo apt install coq
```

Now, open the Emacs terminal by executing the command 'emacs' in the terminal. From this point onwards, we have to use some basic Emacs commands to use Proof General and Coq. This is the reason why I recommended the web-based jsCoq over Proof General for working with Coq. In this way, some of the complexities associated with Emacs can be avoided while handling the intricacies of Coq. However, notice that the Coq syntax used is the same in both these environments.

Once the Emacs terminal is opened, press 'Ctrl+x' and then 'Ctrl+f' to open a file in Emacs. When prompted, give a name to the Coq file in which your Coq source code should be saved. In this example, I have used the file name 'coq1.v'. Notice that Coq files have a '.v' extension. Now, you will see Proof General coming into action. Figure 4 shows the window of Proof General displayed when a Coq file is opened.

Later, a file named 'coq1.v' will be opened in the Emacs terminal. In the file, enter the following line of code: '*Print nat.*' to print the definition of 'nat'. Keep in mind that Coq is case-sensitive, and the dot (.) after the code '*nat*' is mandatory. Now, on the Emacs terminal, press 'Ctrl+c' and then 'Ctrl+n' to run the next step in the Coq program. Since there is just one line in our Coq program, that line will be executed, and the output will be displayed. Figure 5 shows the Emacs terminal when this simple Coq program is executed. The same line of code can also be executed in the web-based environment jsCoq.

The following Emacs commands are also useful for processing Coq programs. To go to the previous step in a Coq program, press 'Ctrl+c' and then 'Ctrl+u'. Press 'Ctrl+x' and then 'Ctrl+s' to save the changes made to the current Coq program being edited. Finally, press 'Ctrl+x' and then 'Ctrl+c'

Figure 4: Proof General window of Coq

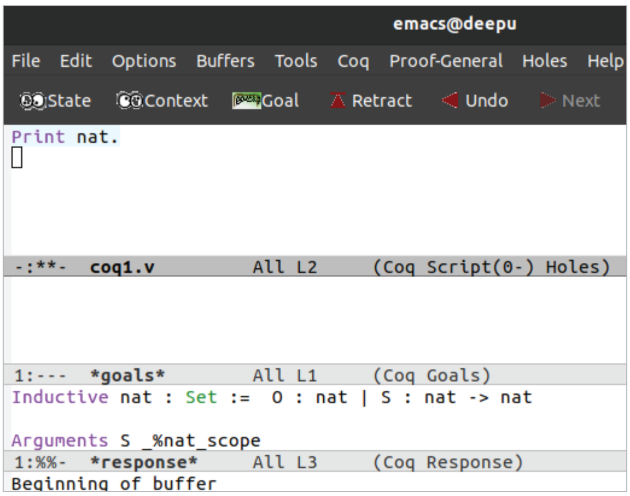

Figure 5: A simple Coq program

to close the Emacs terminal. After executing this command, type 'yes' when prompted to quit.

But what is the meaning of the output shown in Figure 5, which is obtained when the single line of Coq code got executed? The line of code on execution prints the definition of natural numbers (the set of numbers 0, 1, 2, …, denoted by the symbol N). Notice that if you are using jsCoq, when you type 'nat' on the environment, it is automatically replaced with the symbol N. From the definition printed on the terminal, we can see that 0 is defined as a natural number and the successor